

Total No. of Questions : 8]

SEAT No. :

P-495

[Total No. of Pages : 2

[6003]-716

T.E. (I.T)

Software Modeling and Design

(2019 Pattern) (Semester-II) (314454D) (Elective-II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8 .
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Draw Activity Diagram using Swim lanes for given scenario.
“A popular e-commerce website application serves customers to purchase items. Customer can select item by using ‘search item’ facility and add it in the cart. The cart total is calculated, and customer can pay bill by using various payment methods.” [8]
- b) Explain state diagram? Draw and discuss notations used for State Machine, Triggers and Ports, Transitions and conditions, Initial and Final State. [9]

OR

- Q2)** a) Consider typical scenarios of College’s Library Management System, Identify events performed by different users for the system, and Draw Sequence Diagram. [8]
- b) What is interaction and behavior modeling? list different diagram used for this modeling? Draw components used in 2 diagrams. [9]
- Q3)** a) Discuss what is object relational system ? How it is used in designing table class mapping. [9]
- b) Bank offers credit card and debit card to the account holders. Consider different activities to be performed by card holder and draw Class diagram with association. Map Class to tables. Draw schema diagram. [9]

OR

P.T.O.

- Q4)** a) Draw Component and Deployment Diagram for Car rental system. [6]
b) Explain OCL with example. [6]
c) What are the points one should consider while designing the Access Layer Classes [6]

- Q5)** a) How pure fabrication can solve issues related to high cohesion and low coupling? Explain with the help of Sales man as the scenario. [8]
b) Explain Façade Behavioral Patterns. Write in detail any 2 Façade Behavioral Patterns. [9]

OR

- Q6)** a) Apply strategy design pattern to the following and draw the class diagram. A company has many employees. Each employee has a name and a performance index in the range of 1 to 5. When the index is 2 the increment is 10 percent of the previous year salary, 3 the increment is 15 percent of the previous year salary, 4 the increment is 20 percent of the previous year salary. Indicate roll of each class in the class diagram. [8]
b) Explain GOF design patterns. [9]

- Q7)** a) What is real time software architecture? Explain the important characteristics of real time software architecture. [9]
b) What is component based architecture? State different ways of representing component. [9]

OR

- Q8)** a) Describe Quality attributes in architecture design. [8]
b) Explain dynamic view of software architecture with the help of example. [6]
c) Explain the important characteristics of real time software architecture. [4]

